

The constraints can be written as  $Ax = b$  with

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 0 & -1 & 0 \\ 2 & 1 & 0 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}.$$

The system is compatible as

$$x = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

is feasible.

In principle, in the presence of three constraints, we should select 3 variables out of the 4 to be basic variables. But for each of the 4 possibilities, the corresponding basic matrix is not invertible, precluding from eliminating the constraints. The reason is that the constraint matrix is not of full rank. The third constraint happens to be the sum of the two other constraints. It is therefore redundant and must be removed. We obtain the system with the following data:

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 0 & -1 & 0 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

As we have two constraints, we can eliminate two variables. There are 6 ways to do it. But because  $x_1$  cannot be in the basis, we need to select two variables out of three. And there are three ways to do it.

1.  $x_2$  and  $x_3$  in the basis. The basic matrix is obtained by extracting columns 2 and 3 from  $A$ :

$$B = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}$$

which is invertible and happens to be equal to its own inverse:

$$B^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}.$$

In order to apply the generic formula

$$AP = (B \quad N),$$

we need to permute the columns of the matrix so that columns 2 and 3 become the two first columns of the permuted matrix. We use the following permutation matrix,

$$P = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

so that  $PP^T = I$  and

$$AP = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & -1 & 1 & 0 \end{pmatrix}.$$

The constraints can now be written as

$$Ax = AP(P^T x) = Bx_B + Nx_N = b,$$

and we obtain

$$\begin{aligned} \mathbf{x}_B &= \begin{pmatrix} x_2 \\ x_3 \end{pmatrix} = B^{-1}(\mathbf{b} - N\mathbf{x}_n) \\ &= \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix} \left( \begin{pmatrix} 1 \\ -1 \end{pmatrix} - \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_4 \end{pmatrix} \right) \\ &= \begin{pmatrix} -2x_1 - x_4 \\ x_1 + 1 \end{pmatrix}. \end{aligned}$$

The basic variables are  $x_2$  and  $x_3$ , and the non basic variables are  $x_1$  and  $x_4$ .

2.  $x_2$  and  $x_4$  in the basis. The basic matrix is obtained by extracting columns 2 and 4 from A:

$$B = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}.$$

This matrix is not invertible, and the elimination of constraints cannot be performed.

3.  $x_3$  and  $x_4$  in the basis. The basic matrix is obtained by extracting columns 2 and 4 from A:

$$B = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}.$$

It is invertible, and

$$B^{-1} = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}.$$

In order to apply the generic formula

$$AP = (B \quad N),$$

we need to permute the columns of the matrix so that columns 3 and 4 become the two first columns of the permuted matrix. We use the following permutation matrix,

$$P = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix},$$

so that  $PP^T = I$  and

$$AP = \begin{pmatrix} 1 & 1 & 1 & 1 \\ -1 & 0 & 1 & 0 \end{pmatrix}.$$

The constraints can now be written as

$$A\mathbf{x} = AP(P^T\mathbf{x}) = B\mathbf{x}_B + N\mathbf{x}_N = \mathbf{b},$$

and we obtain

$$\begin{aligned} \mathbf{x}_B &= \begin{pmatrix} x_3 \\ x_4 \end{pmatrix} = B^{-1}(\mathbf{b} - N\mathbf{x}_n) \\ &= \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix} \left( \begin{pmatrix} 1 \\ -1 \end{pmatrix} - \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right) \\ &= \begin{pmatrix} 1 + x_1 \\ -2x_1 - x_2 \end{pmatrix}. \end{aligned}$$

The basic variables are  $x_3$  and  $x_4$ , and the non basic variables are  $x_1$  and  $x_2$ .